

Performance of machine learning models in drone detection using optical features

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Abstract—The development of an efficient Anti-Drone System (ADS) is necessary due to the increasing demand caused by the rapid rise in privacy and security risks associated with drones. These risks have become more frequent due to the widespread development and adoption of drone technology. The crucial component required to develop an efficient ADS is the detection unit/model. Several detection models have been developed. However, primarily focused on highly complex deep learning models to detect drones in unauthorized areas using optical features. This work focuses on low-complex machine learning (ML) models, namely logistic regression (LR), support vector machines (SVM), and random forest (RF) for drone detection. In this work, one of the crucial discriminative features of an image namely Histogram of Oriented Gradients (HOG) is considered for drone detection. A comparative evaluation of the three machine learning models is performed in terms of detection accuracy, receiver operating characteristics (ROC/AUC) curves, and computational complexity. The analysis shows that the overall accuracy and inference complexity of the RF model outperforms the other two models, achieving an accuracy of 87.5% on the test dataset, compared to 84.5% and 81.2% for LR and SVM models, respectively. In terms of training complexity, LR is efficient. This confirms that RF model can be an efficient detection model for ADS due to lower inference complexity and low test variance.

Index Terms—Anti drone Systems, Drone detection, Inference Complexity, HOG features, Machine learning models, Receiver operating characteristics.

I. INTRODUCTION

Privacy and security issues have become more prevalent as a result of the widespread adoption and rapid development of drone technology in recent times. While drones have highly beneficial uses for recreational, commercial and military applications, their misuse poses potential threats, including smuggling, unauthorized surveillance, and disruption of infrastructure. Hence there have been equal efforts to develop highly robust and efficient Anti-Drone Systems (ADS) [1]. The main components of an ADS are detection, localization/tracking, and neutralization. The development of an efficient drone detection unit that can correctly classify the image with low false alarm rate is of paramount interest to the scientific community, as it is crucial for a reliable ADS. Methods that involve radar, radio-frequency and acoustic-based techniques have been traditionally used in drone detection [2]. However, each method has its own pros and cons due to limitations, including high costs, detection range, susceptibility to environmental noise, and inefficiency to detect stealthy and small drones. Among these methods, optical-based methods,

particularly those that leverage computer vision and machine learning, have emerged as promising alternatives due to their ability to analyse visual information. These approaches rely on extracting discriminative features from drone images.

In the literature drone detection has increasingly adopted ML techniques due to their adaptability [3]. The authors in [4], [5] have applied manual feature extraction methods such as local binary patterns (LBP), histogram of oriented gradients (HOG), and gray-level co-occurrence matrix (GLCM) for drone detection. Further studies found that LBP achieved lower accuracy than HOG, while GLCM demonstrated promising results but required extensive preprocessing [6]. Coupling these techniques with low complexity machine learning (ML) models such as logistic regression (LR), support vector machines (SVM), and random forest (RF) has demonstrated their potential in detecting drones. A study in [7] indicated that SVM performs well in classifying drones from other aerial objects, particularly when used in conjunction with HOG features. However, its major drawback is sensitivity to noise in the image. The authors in [8] have found that ensemble learning methods are effective in distinguishing drones with low testing variance and can classify drones and birds due to their ensemble learning approach. Deep learning-based feature extraction techniques have received wide adoption because of their outstanding performance [9], [10]. The authors in [11] have also focused on CNN-based classifiers due to the availability of large-scale UAV audio datasets for drone identification. However, their drawback lies in their higher computational complexity, which lies in their need for extremely large data and operations involved. This inherently makes their deployment quite costly, hence techniques like HOG become preferable [12], [13]. Hence, this study focuses on the binary classification (detection) performance of three low complex machine learning models: LR, SVM, and RF. These models were chosen on the basis of the balance that they offer between computational efficiency and predictive performance, making them suitable for drone detection applications in low-cost ADS [9].

The primary objective of this research is to develop a low complex and robust detection model for ADS by comparing the performance and complexity of different ML models such as LR, SVM, RF. The performance in terms of detection accuracy, precision, recall, ROC, and computational efficiency are considered as performance metrics. By analyzing their strengths and limitations, this study aims to provide insights

into the suitability of data driven models for drone detection. The effectiveness of these models is evaluated using drone images obtained from Kaggle database [14]. HOG features are obtained from training dataset to improve the model accuracy. HOG features are considered due to their ability to capture structural information, making them suitable for detecting small objects like micro-drones. They are also computationally efficient and work well with low-complexity models.

The remainder of the paper is structured as follows: Section II presents the problem statement and the methodology for drone detection using HOG features. Section III discusses the performance of low-complexity ML models based on simulation results, followed by the conclusions in Section IV.

II. PROBLEM STATEMENT AND METHODOLOGY

A. Problem Statement

The proposed framework of the drone detection unit using optical features is illustrated in Fig. 1. A dataset containing drone and non-drone images is collected from Kaggle database. The images are preprocessed before the dataset is split into training and testing sets. HOG features are extracted from all drone images and non-drone images. The extracted features are used as inputs to train considered ML models before they are evaluated. The model can then be used to detect a drone in an image which is captured in real-time.

The problem of finding a drone in an unauthorized area using optical features can be considered as a binary classification problem. Hence, the goal is to predict whether a drone is present in an image based on its HOG features.

B. Methodology

Let I_d be a drone image from training dataset, whose HOG Feature Vector ($\mathbf{F}$) is a compact representation of the drone image that captures gradient information. It can be computed as:

$$F = \text{Concatenate}(\text{Norm}(\text{Hist}(\text{Grad}(I_d)))).$$

Logistic Regression: Let the training dataset $\{(\mathbf{F}_i, y_i)\}_{i=1}^N$, where $\mathbf{F}_i \in \mathbb{R}^d$ is the HOG feature vector of size d and $y_i \in \{0, 1\}$ is the binary-class label. Then, the hypothesis function $h(\mathbf{w}, b, F)$ predicts the probability that a drone is present in the image, given the feature vector $\mathbf{F}$ [15],

$$h(\mathbf{w}, b, F) = \frac{1}{1 + e^{-(\mathbf{w}^T F + b)}}, \quad (1)$$

where, $\mathbf{w}$ and b are the weight vector and the bias, respectively.

The decision logic to discriminate the hypothesis based on Logit function ($h(\mathbf{w}, b, F)$) is given by:

$$\hat{y} = \begin{cases} 1 & \text{if } h(\mathbf{w}, b, F) \geq 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

Where, 1 and 0 represents the drone detected and not-detected in a given image, respectively.

The optimal parameters for Logit function can be calculated by minimizing the log-loss (cost) function using Batch Gradient decent algorithm, which can be expressed as:

$$\min_{\mathbf{w}, b} \left[-\frac{1}{N} \sum_{i=1}^N \left(y_i \log(h(\mathbf{w}, b, F_i)) + (1 - y_i) \log(1 - h(\mathbf{w}, b, F_i)) \right) + \lambda \|\mathbf{w}\|^2 \right].$$

Where, N represents the number of training samples, λ denotes regularization parameter, and $(\|\mathbf{w}\|^2)$ is the $L2$ norm of the weight vector.

Support Vector Machine: The hypothesis function to predict the class using feature vector ($\mathbf{F}$) is [13],

$$h(\mathbf{w}, b, F) = \mathbf{w}^T F + b, \quad (2)$$

where, $\mathbf{w}$ and b are the weight vector and the bias, respectively.

The decision logic to discriminate the hypothesis based on Logit function ($h(\mathbf{w}, b, F)$) is given by:

$$\hat{y} = \begin{cases} +1 & \text{if } h(\mathbf{w}, b, F) \geq 0, \\ -1 & \text{otherwise.} \end{cases}$$

Where, +1 represents the drone presence and -1 represents the absence of drone in an image.

The optimal parameters for hypothesis function can be calculated by minimizing the cost function, which can be expressed as:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \cdot \sum_{i=1}^N \max(0, 1 - y_i(\mathbf{w}^T F_i + b)),$$

where, C represents regularization parameter, which is crucial for balancing bias and variance of the SVM model, and $(\max(0, 1 - y_i(\mathbf{w}^T F_i + b)))$ is the hinge loss function.

Random Forest: Given the feature vector F , the hypothesis $h(F)$ predicts the probability of a drone being present in the image as [16]:

$$h(F) = \frac{1}{T} \sum_{t=1}^T f_t(F). \quad (3)$$

Where, T denotes the number of trees used to for Random Forest global prediction. The function $f_t(F)$ represents the individual prediction of t_{th} decision tree. Each tree (t) is trained using bootstrap sampling of N data points.

The hypothesis is classified using the following decision logic:

$$\hat{y} = \begin{cases} 1 & \text{if } h(F) \geq 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

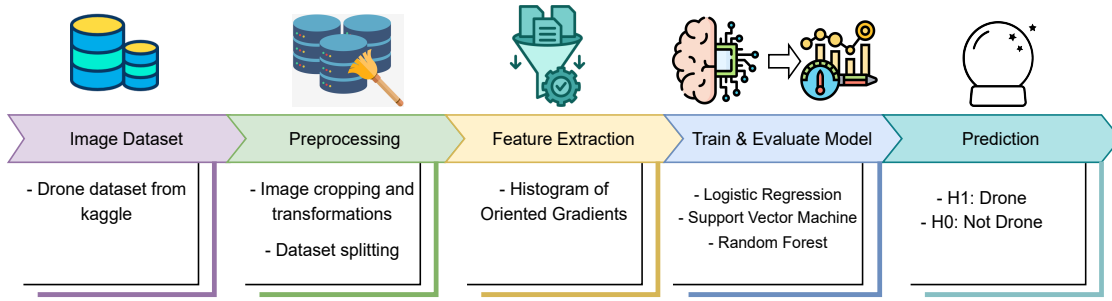


Fig. 1. Block diagram of proposed drone detection method using optical HOG features.

III. SIMULATION RESULTS

The simulations were designed to replicate real-time drone detection scenario. The complete dataset consists of 10,800 images, which are split into training (80%) and testing (20%) sets. The 80% dataset of 8999 images was again divided into training (75%) and validation (5%). Out of these images, 48% belong to the drone class (1), while 51% belong to the non-drone class (0). The non-drone class contains images related to airplane, birds, image with no object, and helicopter. The images were collected from publicly available sources and annotated. Each image was resized to a standard dimension of 128×128 pixels and converted to grayscale to reduce computational complexity. Additionally, the HOG feature extraction technique was applied to generate feature vectors for drone detection. The extracted features were normalized using min-max scaling, given as $F_{\text{norm}} = \frac{F - F_{\min}}{F_{\max} - F_{\min}}$. Where, (F) represents the feature vector, and $F_{\min}$, $F_{\max}$ are the minimum and maximum feature values, respectively. The simulations were conducted using a 5-fold cross-validation approach. Batch gradient descent (BGD) optimization is considered with learning rate (α) of 0.001. The Google Colab is used as an IDE to simulate models, which are written in the Python programming language. The performance evaluation of considered models has been analysed using precision-recall and ROC curves, which are derived from corresponding confusion matrices.

In general, the computational complexity of ML model is dependent on number of images for training the model $(N = 8999)$, size of the feature vector $(d = 8100)$, and the model subsystems (number of decision trees $(T = 100)$). Table. I provides computational complexity of LR, SVM, and RF models and its comparison with Convolution neural networks (CNN). For a given balanced dataset of 8999 images, LR (Batch GD) and SVM show better choices with respective to training complexity, $O(8999 \times 8100 \times i)$ per i number of iterations. And for one iteration, it is equal to $O(8999 \times 8100) = O(7.28 \times 10^7)$. But, the value of i depends on α which updates the weight value during batch gradient decent (BGD) optimization. Whereas, RF has a significantly higher training complexity $O(T \times N \log N \times \sqrt{d}) \cong O(9.2 \times 10^8)$, and has lower inference complexity (i.e., it can predict quicker when unseen data is given) when compared to LR and SVM. It is

calculated as $O(T \log N) = O(100 \times \log 8999) \cong O(1300)$. Because deep learning techniques such as CNN have to process several layers (convolutions, pooling, and fully connected layers), their training and inference complexity is much higher than considered low-complexity ML models [10].

The confusion matrices in Fig. 2 and 3 summarize the performance of each model at classifying the input image as either Drone (1) or Not-Drone (0). The blue, red and green plots represent LR, SVM, and RF models, respectively. The logistic regression model gave 197 true positives (TP), 208 true negatives (TN), 34 false positives (FP), and 40 false negatives (FN). Out of 497 testing images, 405 were correctly classified, while 74 were misclassified. The slightly high number of false negatives (40) would affect this model's performance in terms of recall. These are instances of actual drones that the model failed to detect, which would be undesirable in an ADS. In the case of SVM, 389 images were correctly classified, while 90 images were incorrectly classified as either false positives or false negatives. The model shows a descent classification ability, however its performance is poor compared to LR in terms of recall (shown by the higher number of false negatives). In the case of RF-based detection, the model yielded the best testing results, correctly classifying 419 images and misclassifying only 60. Among the three models, it gave the highest number of true positives and least false negatives. In conclusion, RF exhibited the best overall performance in terms of correct classification.

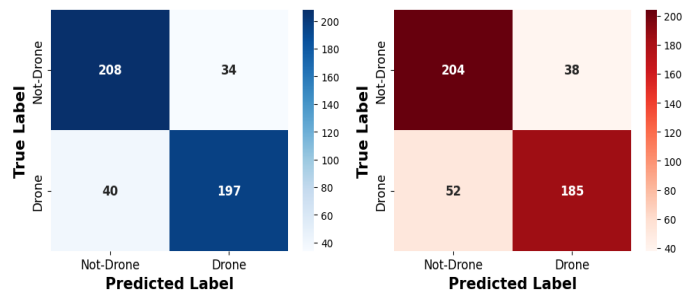


Fig. 2. Confusion matrix of LR (blue) and SVM (red) models.

Table. II provides summary of the performance metrics. RF yields the highest testing accuracy of 87%. It also achieved the highest recall value of 91%.

TABLE I
 COMPUTATIONAL COMPLEXITY COMPARISON OF CONSIDERED ML MODELS.

ML Model	Training Complexity	Inference/Testing Complexity	Observations
LR (BGD)	$O(N \cdot d \cdot i) = O(7.28 \times 10^7)$ per iteration	$O(d) = O(8100)$	Efficient
SVM	$O(N \cdot d) = O(7.28 \times 10^7)$	$O(d) = O(8100)$	Feasible
RF	$O(T \cdot \sqrt{d} \cdot N \log N) \approx O(9.2 \times 10^8)$	$O(T \log N) \approx O(1300)$	Efficient once trained

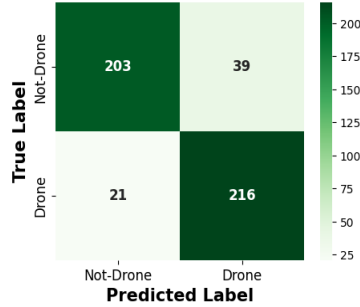


Fig. 3. Confusion matrix of RF model.

TABLE II
 CLASSIFICATION METRICS FOR THE MODELS LR, SVM, AND RF.

Class	Metric	LR	SVM	RF
Not-Drone	Precision	0.84	0.80	0.91
	Recall	0.86	0.84	0.84
	F1-Score	0.85	0.82	0.87
Drone	Precision	0.85	0.83	0.85
	Recall	0.83	0.78	0.91
	F1-Score	0.84	0.80	0.88
Overall	Accuracy	0.85	0.81	0.87
	Macro Avg	0.85	0.81	0.88
	Weighted Avg	0.85	0.81	0.88

Fig. 4 illustrates the precision against recall of the models. Recall gives how many actual drones were correctly detected, while precision gives how many predicted drones are actually drones. The area under the precision-recall curve gives the average precision (AP). From the figure, it is clear that the AP is highest for RF (0.94), followed by LR (0.91) and SVM (0.88). This graph can indicate that in situations, where highly dependable drone detection is required, RF can be applied.

The ROC curve analysis in Fig. 5 reveals the comparative performance of LR (blue), SVM (red), and RF (green) models by varying the threshold values in classifying drone features in images. The RF model (green) demonstrates the optimal classification ability. The area under its ROC curve namely AUC value of 0.95 indicates its effectiveness in classifying drone binary classes. This suggests that RF is the most reliable model for the considered dataset when we compared with other two models. The LR model (blue) with an AUC of 0.92 performs well but slightly lags behind RF. While it is a simpler model, its inference complexity is high. The SVM model (red), though still effective, has the lowest AUC at 0.90, indicating that model performance is poor compared to LR and RF. Overall, Random Forest emerges as the best model, followed

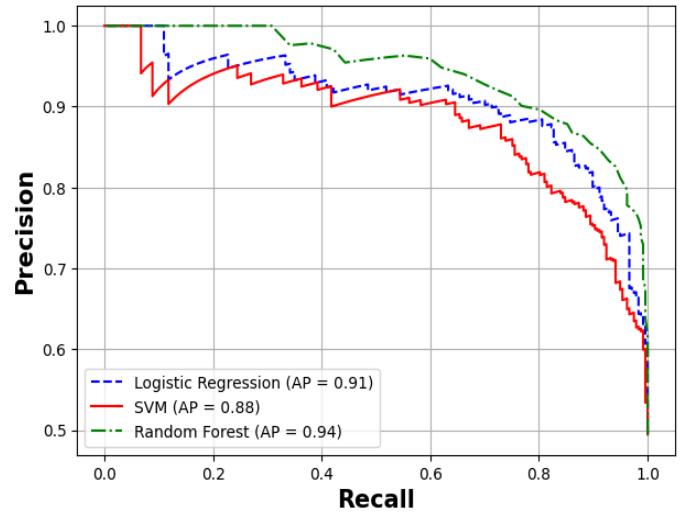


Fig. 4. Performance of considered ML models in terms of Precision against Recall.

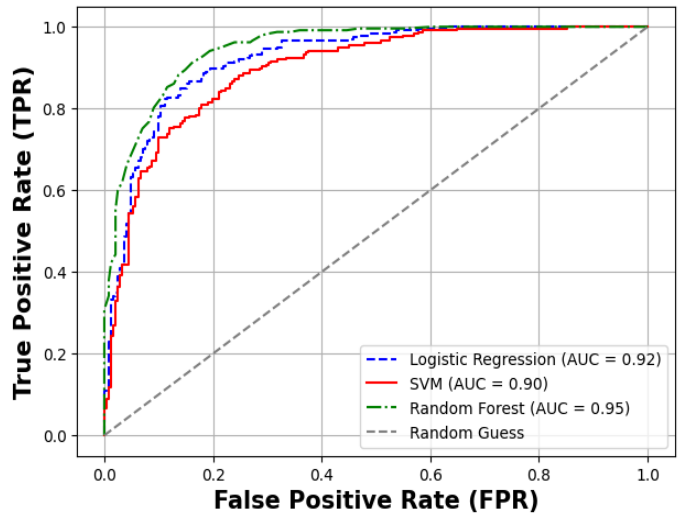


Fig. 5. ROC curves of three low complex ML models.

by Logistic Regression, while SVM is the poor performer.

IV. CONCLUSIONS

This paper proposes a drone detection method using optical features based on low-complexity ML models. A comparative analysis of LR, SVM, and RF has been conducted. The models were trained on a balanced dataset using HOG features. The evaluation results demonstrate that these models

exhibit reasonable classification performance. However, their effectiveness varies significantly in terms of detection accuracy and inference complexity. Among them, RF emerges as the suitable choice for drone detection for ADS. From the results, the accuracy of RF is 87% followed by LR 85% and SVM 81%. The precision-recall curves further reinforce these findings. Random Forest achieves the highest Average Precision (AP = 0.94), indicating an optimal trade-off between precision and recall. LR with an AP of 0.91, performs well but exhibits fluctuations in precision. SVM, with an AP of 0.88, has the weakest performance. The ROC curve analysis confirms these observations, with RF achieving the highest AUC of 0.95, followed by LR of 0.92 and SVM of 0.90. The higher AUC signifies better RF model performance. Overall, RF consistently demonstrates superior classification ability and low inference complexity making it the best choice for detection unit of ADS. Future work could focus on enhancing performance through hyper-parameter tuning. Additionally, ensemble techniques incorporating multiple optical features could further improve model training and overall classification accuracy.

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